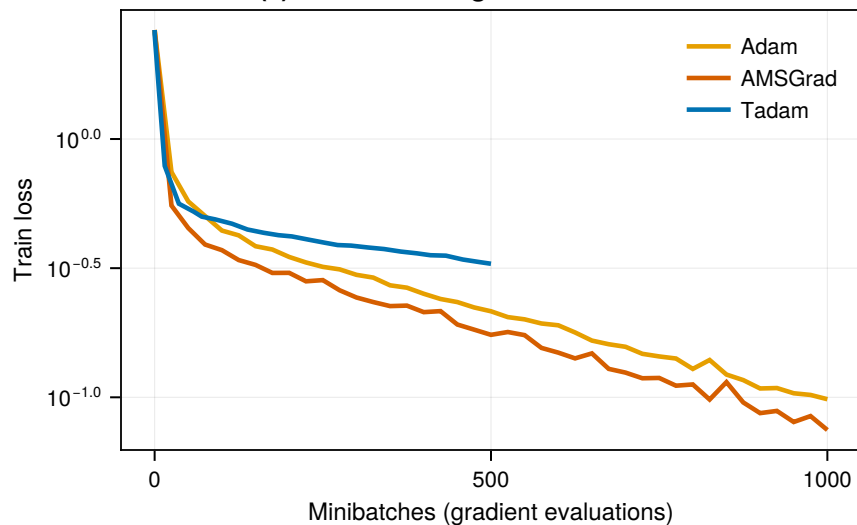
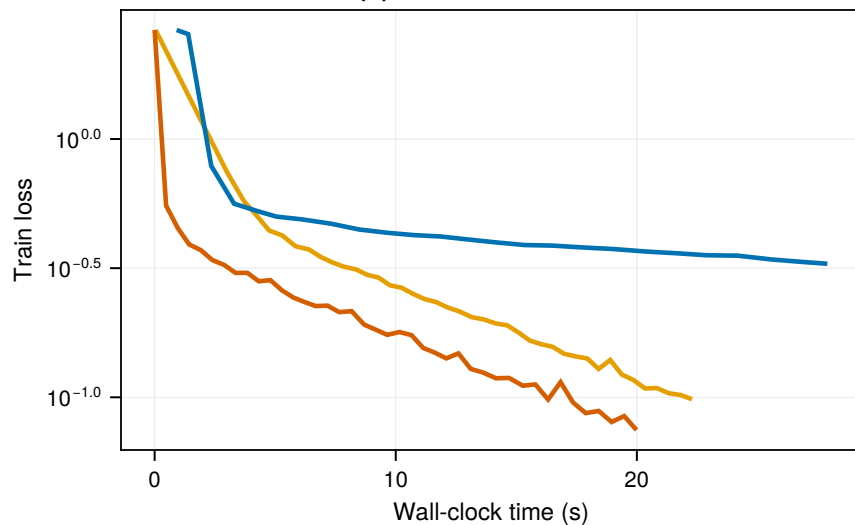
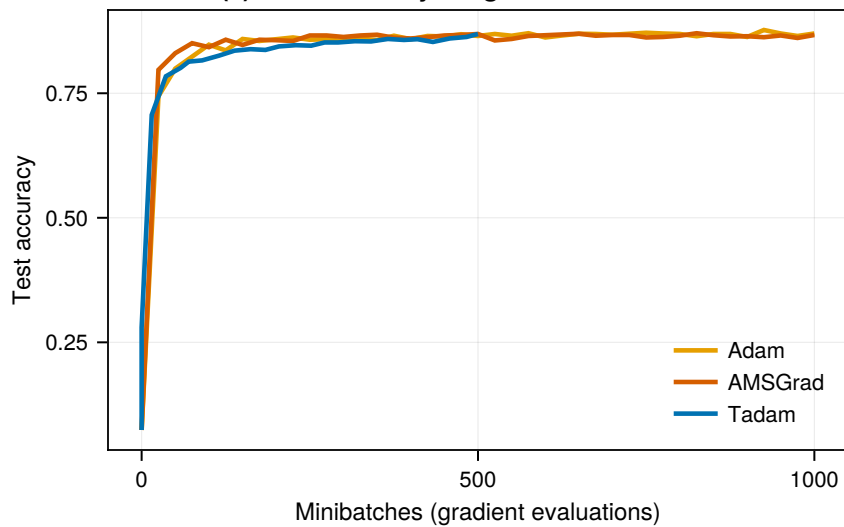
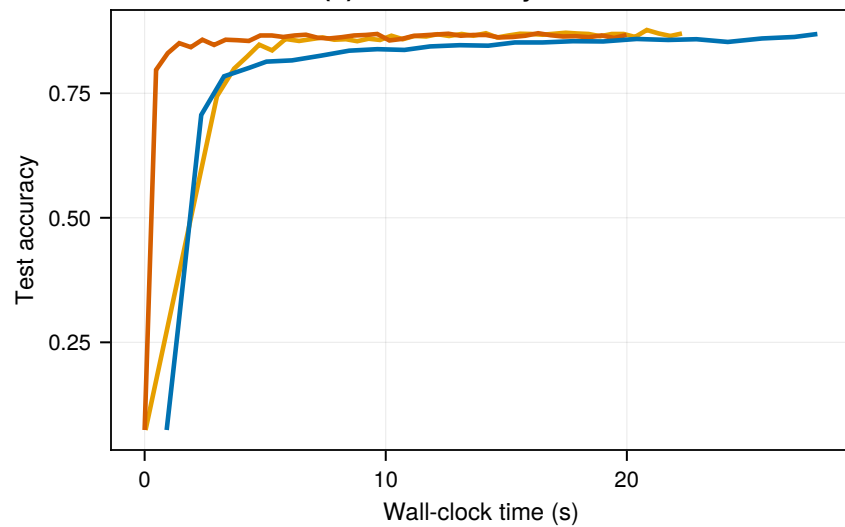
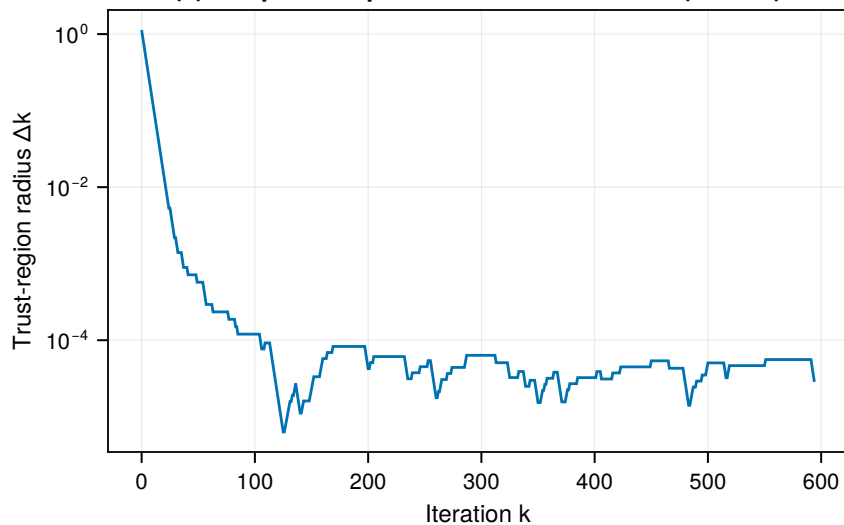
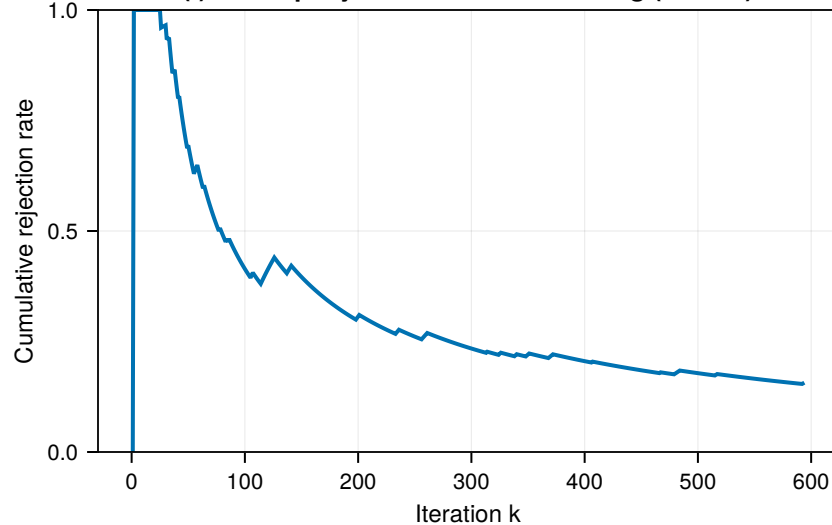


(a) Train loss vs. gradient evaluations**(b) Train loss vs. time****(c) Test accuracy vs. gradient evaluations****(d) Test accuracy vs. time****(e) Adaptive step size: Δk over iterations (Tadam)****(f) TR step rejection rate over training (Tadam)**

Fashion-MNIST ($N_{\text{train}}=10000$, $N_{\text{test}}=2000$, batch=512). MLP: 784 $\rightarrow$ 512 $\rightarrow$ 256 $\rightarrow$ 10. Adam & AMSGrad use $\text{lr}=0.0003$; Tadam adapts Δ automatically.
 Panels (e–f): Tadam internal diagnostics showing self-tuning behaviour.